Design FMEA (qualitative, simulation-signal based) — t5_r1_flash

Doc No. VBF-T5R1FLASH-DFMEA-01 · Rev A (2026-08-25) · Qualitative caliber: severity/occurrence rated from simulation signal magnitude vs threshold; S/O ∊ high/medium/low; RPN = simplified S×O matrix (annotated qualitative).

Failure mode	Failure cause	Simulation signal (detectability)	Severity	Occurrence	Design-side mitigation
---	---	---	---	---	---
Negative-electrode Li plating (fast charge)	anode potential < 0 V at 4C: deep-charge lithiation + electrolyte depletion	anode_potential_vmin = **+14.7 mV** (margin vs 0)	low (caught before onset; plating risk currently absent)	low	N/P 1.92 (180 µm negative), negative ε 0.32, D 1e-9/σ 6 (transport), sep ε 0.55 — R4→R9 chain −41→+15 mV
Thermal runaway risk (4C charge)	heat generation > dissipation at 4C	T_max 327.55 K vs 333.15 K limit (margin 5.6 K)	high (consequence if triggered)	low	h=100 W/m²K cooling; margin 5.6 K; transport gains cut ohmic heat
Electrolyte oxidative decomposition (4.7 V window)	high-voltage oxidation at 4.7 V cathode	Stage 2 funnel: additive set FEC/VC/PES/DTD passed elimination lines (HOMO < −6 eV threshold; HOMO −12.8…−12.0 eV); true DFT endorsement skipped (real_compute=false)	medium	low	4.7 V-class LNMO system + film-forming additives (SEI/CEI)
Insufficient capacity / energy	electrode utilization loss	1C capacity 8.26 Ah, ED 597.9 Wh/kg vs 500.94 (margin 97 Wh/kg)	medium	low	positive-limited design; stoich window 0.9→0.3
Electrolyte transport degradation at low T	conductivity drop	not simulated (low-T out of scope)	medium	not evaluated	N/A (beyond simulation boundary)

Conclusion: highest-risk items are high-consequence/low-occurrence (thermal) — mitigated by 5.6 K T_max margin and plating-free anode (+14.7 mV margin). All identified failure modes have design-side mitigations implemented in LNMO-R6. Complete FMEA incl. process/supplier failure modes: **N/A (beyond pure simulation boundary)**.